Neurophysiological Correlates of Top-Down Alpha Oscillatory Modulation and Neural Entrainment as Mechanisms of Attention in Children Cochlear Implant Patients

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Abstract:

Recruitment and modulation of neural networks facilitates and suppresses particular sensorineural processing mechanisms, with extensive recent audiological research focusing on selective attention. A mechanism of unconscious automaticity, or minimal effort voluntarily, in normal hearing individuals, is disrupted and cognitively taxing in those with hearing impairments. Cochlear implants succeed in providing absolute hearing improvement providing greater functionality and quality of life, however research is inconclusive as to how moderate to severe sensorineural hearing loss specifically alters these mechanisms and how cochlear implants lack the ability to compensate. We consider and examine how top-down modulation and phase sequential modulation may influence enhancement and suppression of attention and correlate to self perceived speech comprehension and cognitive demand in CI patients. Accordingly, we will record high density electroencephalograms (EEG) while CI users listen/watch episodes of the TV show, Malcolm in the Middle, in a cocktail party effect paradigm. Multi-babble talk will project from masking speakers in three randomized conditions of left, right or collocated, to parallel an environment of multiple sound sources. A quiet condition will also be executed as a baseline and to contrast the high perceptual load conditions. Subsequent self perceived cognitive demand and speech/content comprehension tests of trials will be performed. Follow-up to completion of all trials a visuospatial memory test will be executed to measure individual working memory capacity. Processing of continuous EEG recordings will be used to quantify neural entrainment (i.e., TRFs, temporal response functions) to the attend, episode, and ignore, babble talk, speech envelopes and to measure magnitude and localization of alpha oscillations, 8–12 Hz, associated with attentional effort.

Introduction:

A cochlear implant (CI), novelly a “treatment of choice” for moderate to severe sensorineural hearing loss, generates signals internally and transmits them directly to the auditory nerve fibers within the cochlea, bypassing damaged parts of the auditory pathway to be recognized as sound (National Institute on Deafness and other Communication Disorders, 2021). Post-implantation studies have reported gains in speech perception scores and subjective functional improvements (Forli F et al., 2017). However, these devices are limited in their ability to generate high-quality

hearing across all perceptual conditions and interindividual variability. Advantageous from the specialization of the mature brain, the plasticity of children's brain's provides great potential for anatomical and functional flexibility. Numerous studies have investigated the critical period of cortical plasticity where children who receive a cochlear implant before 18 months of age and receive early therapeutic training develop language acquisition skills relative to children with normal hearing (Campbell et al., 2014) (Arras et al., 2022) (Arras et al., 2023). Nonetheless, inconclusive empirical evidence exists regarding the ability of children with CIs to function in complex auditory environments, likewise to adult CI research.

The cocktail-party effect, coined by Colin Cherry in 1953, encapsulates the phenomenon of attentional dispersion where in normal hearing individuals can discern a single speech stream among the many competing speech streams present in a noisy environment effortlessly directing cognitive attention (Cherry, 1995). The technological design of speech processing systems within CIs has improved in recent years, however, the improvements are most evident in conditions of quiet and single streams where perceptual load is low (Lesica et al., 2021). As a result of CI mechanics, speech sounds are transformed in the brain with a greater degree of degradation, taxing demands on the brain's cognitive resources to interpret the degraded signals and filter out background noise. Strain on the cognitive resources of CI users frequently results in a decrease in motivation and effort, thus resulting in a decrease in quality of life. Analytical methods for the locus of auditory attentional selection both in normal hearing individuals and cochlear implant patients most frequent artificial, non-speech stimuli in relatively uncluttered scenes (Miller, 2013). Furthermore, most studies of selective attention fail to evaluate speech output's unique attributes as a temporally extended and multifaceted sound.

Lateralized alpha cortical oscillations have emerged in recent research as a general marker of facilitation of a top-down cortical gating mechanism for attentional modulation. Prior research has demonstrated that these oscillatory frequencies not only serve as indicators of cognitive effort, but also exhibit sensitivity to neural entrainment (Dimitrijevic et al., 2019) (Paul et al., 2020) (Paul et al., 2021) (Bowen Xiu et al., 2022). In hemispheric sensory processing, frequencies of cortical oscillations are recruited differently: alpha (8-12 Hz) and beta (13-30 Hz) oscillations tend to decrease, whereas theta oscillations (4-8 Hz) and gamma oscillations (30-150 Hz) tend to increase when attending to stimuli (Payne & Sekuler, 2014). Alpha desynchrony in a hemisphere is associated with perceptual alertness and attention in the contralateral hemifield, such as the attended stimuli in paradigms. Whereas, a greater degree of alpha synchrony in one hemisphere decreases neuronal excitability in localized neurons responsible for processing stimuli, associating reduced attention in the contralateral hemifield, associating to distracting stimuli in cocktail party paradigms. Placing the neural circuitry normally responsible for sensory interpretation in some form of inhibited or decreased receptivity state suppresses or entirely limits processing of the competing or distracting information. Alpha frequency cortical gating has been demonstrated in the visual and somatosensory domains (Haegens et al., 2011). A

contralateral alpha suppression has been identified at posterior sites associated with visuospatial attention, which has been attributed to from higher order fiber frontal areas like the IFG and delivering to parieto-occipital areas. (Wang et al., 2016). Despite growing research, the mechanism of alpha suppression and its temporal and spatial activity is unconcluded in auditory perception, yet it is proposed that visuospatial attention and auditory spatial attention share a common basis of mechanics.

An additional key mechanism by which the brain bypasses this multiple-talker disarray is neural entrainment, often referred to as phase coding or speech tracking. Prioritizing relevancy, neural representations of sensory information are structured by features of the auditory input temporally aligning with low frequency (1-8 Hz) cortical activities. There has been evidence in the literature that attended speech has a preferential neural tracking over ignored stimuli in temporal brain areas superior and proximal to the primary auditory area, Heschl's Gyrus (Mesgarani & Chang, 2012) (O'Sullivan et al., 2014). It is currently possible to unravel the selected attentional streams of listeners utilizing temporal response functions (TRFs) where features of streams and EEG cortical responses are mapped to discriminate attended and ignored sources (Fu et al., 2018). A facilitated speech in noise paradigm measured temporal response functions and found deficits in early-stage separation of cortical speech representations with controls showing stream separation at a 150 ms mark and CI patients at a 250 ms mark (Paul et al., 2020).

In the following study we will replicate naturalistic high perceptual load environments, using continuous and naturalistic audiovisual stimuli with spatial allocation of multi-talker background noise. The study aims to specifically address evidence for the role of oscillatory modulation and the neural entrainment of multifaceted features as selective attention mechanisms in normal hearing and CI individuals. A mechanistic link between spatial and selective attention in a "cocktail party" could be established by hypothesizing that selective attention recruits lateralization of alpha activity at perisylvian and posterior parieto-occipital cortical sites predicting the increase in attended signals by frequency entrainment**.** Source analysis will localize electrical desynchronization and synchronization of alpha-band oscillations with expectations that the irrelevant ignored task stimuli will produce relative ipsilateral alpha enhancement and contralateral alpha suppression in primary auditory areas and sensory integrative processing areas, with ROI’s of left and right superior temporal gyrus, transverse temporal cortex and parieto-occipital areas. With temporal response functions (TRF’s) analysis, we expect alpha-band synchrony to associated irrelevant hemifields and alpha-band desynchrony to attend hemifields as indicators of a gain of low frequency phase locking in auditory cortical activity to attended speech streams. Additionally, we hypothesize a quantifiable relation of modalities to the subjective self-reported level of cognitive demand, self-perceived comprehension, and individual working memory capacity.

Materials and Methods:

Participants:

Children CI users will be recruited from the Hospital for Sick Children, Department of Otolaryngology. The inclusion criteria for participants will include children CI users of unilateral or bilateral hearing loss between the ages of 6 and 16 with at least one year of experience with their implants. Additionally, there will be an age matched control group of participants with typical development associated with normal hearing participated in the study. We intend to test 15 participants in each condition group. Participants will be of native or bilingual fluency in the English language in consideration of English presented stimuli. Prospective participants were not excluded based on their hearing loss etiology, their hearing loss onset age, their performance on clinical speech perception tests, or their implantation side. Audiograms will be evaluated using a pure tone audiometry machine prior to any experimental conditions to identify the hearing thresholds of individuals and deem them applicable to the normal hearing (0-25 Hz) or cochlear implant conditions (>40 Hz without CI, 0-25 Hz with CI). Pure tones are presented as an air conduction test, through headphones, presenting bursts of pure tones at audiometric frequencies of 250, 500, 1000, 2000, 4000 and 6000 Hz. Participant data will be excluded due to poor electroencephalogram (EEG) recording quality. Demographics of age, sex, implant side, CI device and duration of use, and etiology, will be provided.

In accordance with the Tri-Council Policy Statement: Ethical Conduct for Human Research approval of methods and protocols were directed by the Sunnybrook Health Sciences Centre Research Ethics Board (REB #2028). Parents and/or guardians provided written informed consent on behalf of participants. In exchange for their time and participation, all participants received monetary compensation and reimbursed for parking expenses at the hospital campus.

Listening Environment and Experimental Setup:

The continuous audiovisual paradigm will be conducted in an electrically shielded and sound-attenuated booth with minimal reverberation. Participants will be seated at the center of a circular ring array of eight speakers (0°, ± 45°, ± 90°, ± 135°, and 180°), each located 80 cm away from the center and raised 1 m from the ground. Speaker allocation is relative to the participants frontal position towards the center computer monitor presenting the audiovisual stimuli, placed 0.80 m directly in front of the listener's head, under the speaker located at 0° azimuth. The attended stimuli is presented at 65 dBA and multi-talker babble noise is presented at a level of 65 dBA in the relative conditional speakers as measured at the center of the ring array using the Brüel & Kjær Type 2270-S Sound Level Meter (B&K, Nærum, Denmark). Adobe Premiere Pro and Windows Media Player were used to process and display episode

segments, and the Tucker Davis Technologies (TDT, Alachua, FL, USA) RX8 processor controlled audio presentation of episode and background stimuli.

Stimuli and Procedure:

The stimuli used for the paradigm are 15 minute audio and video segments of episodes from seasons one and two of the television show *Malcolm in the Middle*. Three runs for each condition are performed with the order of runs and conditions randomized for each participant to avoid bias, for a total of 45 minutes per condition and 180 minutes experiment duration. The show was selected based on its features of continuous naturalistic dialogue with little to no music and content of suitable language and interest for children. To further minimize non-speech sounds and replicate a cocktail party scene, the opening theme song sequence, and irrelevant scenes with music, non-verbal distracting background noise, and of no dialogue were removed from episodes. To investigate cognitive demand in complex listening conditions, segments of the show are presented in conditions that vary in multi-talker babble noise levels. The babble noise soundtrack used for the current study was adapted from the four-talker babble noise used in the Quick Speech in Noise (QuickSIN) test (Killion et al., 2004).

The procedure as illustrated in Figure 1, will require participants to attend to the audiovisual stimuli played from the loudspeaker at 0° azimuth located directly in front of the listener. Simultaneously, relative to the four conditions, ignored multi-talker background stimuli originates from surrounding ± 90° speakers or collocated with the attended audio at 0°. Conditions consist of quiet, in which audiovisual stimuli was presented naturally with no multi-talker babble noise, or collocated, where distracting noise is produced from the speaker at 0° of the attended audio and visual stimuli. Additionally, to measure lateralized cortical responses, two paralleled conditions present multi-talker babble noise from the left speaker (-90°) or the right speaker (+ 90°). As a measure of their attention to stimuli, participants will be asked to answer four multiple choice comprehension questions after each episode, two associated with event related content and two associated with content dialogue. As a result, participants averaged ... correct (SD = ...). Self perceived cognitive measures of demand will be administered after each episode, providing a subjective numerical scale of 1-10, with the endpoints on the left and right being “Low” and “Very High,” respectively, to the question: "How mentally demanding was it to maintain attention to the presented show?”. Additionally, participants are asked Yes/No if attention was given up, measuring cognitive capacity.

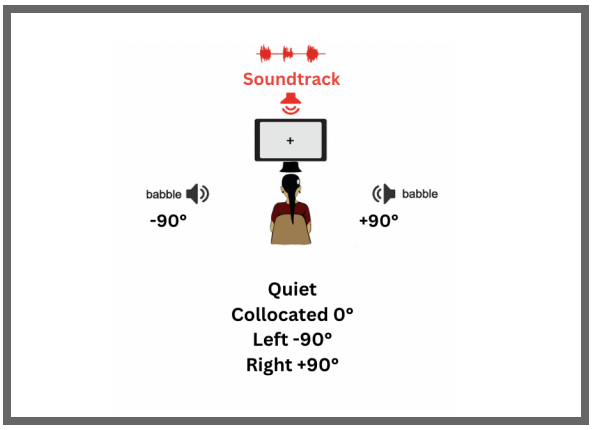


Figure 1. Schematic of audiovisual paradigm. Participants are centered surrounded by an array of speakers with an attended soundtrack of audiovisual stimuli projecting from 0°. Condition dependent distractor speakers alternate multi-talker babble noise from - 90°, + 90° and 0°.

Working Memory Dot Test:

Providing the first evidence in 2009, a causal role for working memory in reducing interference by irrelevant auditory distractors was demonstrated (Dalton et al., 2009). More recently studies investigating the fronto-parietal working memory networks have shown deaf children with CIs may have altered encoding strategies and less efficient rehearsal processes for working memory (Pisoni et al., 2016). To measure relations of interindividual cognitive demand and working memory to a between subject analysis participants will be asked to complete a Python scripted visuo-spatial working memory test. This working memory dot test was developed from a digital adaptation of the Corsi Block Test, frequently used to evaluate conditions such as acquired brain traumas, dementia, Parkinson's, and Alzheimer's diseases (Arce & McMullen, 2021). The Running Dot Working Memory Test presents an encoding phase of a series sequence of 2 to 7 dots, displayed one by one on a 9-squared grid at varying locations. At the immediate retrieval phase, participants are presented the 9-square grid and must digitally recall the sequenced order. The number of different dot displays presented to the participant increases from 2 to 7 dots per trial across 6 experimental trials. The Dot Test and its 6 trials is performed twice, with a forward retrieval condition and a reverse retrieval condition. Participants recall the dot sequence in the order it was presented to them in a forward model (first dot displayed to last dot displayed), and subsequently perform a new set of trials recalling sequences in a reverse order of the sequence presented (last dot displayed to first dot displayed).

Electroencephalogram Recording and Preprocessing:

Electroencephalogram data is continuously recorded at a sampling rate of 2,000 Hz using a 64-channel EEG antiCHamp Brain Products recording system (Brain Products GmbH, Inc., Munich, Germany). EEG caps are fitted according to the 10–20 international system. Typical electrode placement was localized by Cz at the vertex of the skull, as determined by measurements of the midsection junction between nasion-to-inion and tragus-to-tragus. Based on a slightly anterior reference electrode on the midline, separate localizations are performed for digital electrode referencing with the ground electrode located on the midline between the naison and the vertex. The electrodes on the side of the tested ear, ranging from 1 to 4, which overlapped or were within proximity to the CI magnet and coil, will not be used. We will collect individual anatomical landmarks and fiducial points of EEG caps and head measurements of participants during sessions, obtaining them during Brainstorm brain source analysis from the sample dataset, (.pom files). Pom files of digitized head points will be acquired using Polhemus Patriot (Polhemus, Colchester, VT, USA) digitizing all electrodes and fiducials to produce a 3-D map.

MATLAB 2023b (The MathWorks, Natick, MA, USA) will be used to perform the electroencephalographic preprocessing and processing using EEGLAB 2023 (Delorme and Makeig, 2004). After importation, raw EEG recordings are filtered with a Basic Finite Impulse Response bandpass filter from 0.1 to 40 Hz . To maximize pre-processing temporal alignment between audio stimuli and EEG data, an auxiliary channel is used to record audio segments simultaneously with electrode recordings. To optimize alignment accuracy, 5 second duration high frequency sweep sounds were attached to the start and end of episode runs. Considering artifactual noise of the EEG auxiliary channel, trial audio soundtracks were extracted, using Adobe Premiere Pro, and will be aligned offline to the auxiliary audio stream of trials by calculating cross variance and time lag to start. Audio soundtracks will further be extracted down to speech envelopes by calculating the absolute value of the Hilbert transformation of the aligned audio. The speech envelope is downsampled to the sampling rate of EEG recording, 2,000 Hz and will be appended to raw data for future processing. After alignment, appended audio EEG data will be downsampled to 250 Hz in EEGLAB.

As part of the artifact removal process, the raw continuous EEG data will be visually analyzed, removing channels and global EEG sections of high noise duration. Excluding marked epochs of noise, concatenated EEG data will be processed through EEGLAB Independent Component Analysis (ICA) using an Infomax Runica.m algorithm (Pontifex et al., 2017). Visual inspection of ICA component topographies of identified stereotypical physiological and technical artifacts, notably eye blinks, cardiac activity, oculomotor movements, and electrode line noise, were manually removed. Statistical analysis of mean and standard deviation of components corresponding to artifact topographies removed will be correlated for each participant. Removed

EEG channels and channels that are not recorded due to CI proximity will then be reconstructed using the spline interpolation tool on EEGLAB, derived from estimates of neighboring sensors.

Cochlear Implant Artifact Suppression:

In part of cochlear implant speech transduction, electrical stimulations and radio frequency pulsations interrupt electrode recordings. The electrical recordings obtained may consist of CI-induced artifact obscurities, representing poor neurological signal representations and hindering verifiable analysis and conclusive results. Various methods are being explored to reduce artifacts including ICA component analysis, while it remains a challenge to isolate the artifact from the neural response, especially for continuous stimuli of multi-faceted components (Wagner et al., 2018). Generally, complexity of disentangling implant artifacts from neural signals as opposed to physiological artifacts results from the difference in temporality. Artifacts of physiological and technical properties are not temporally aligned to stimuli onset and furthermore are separated components easily deciphered in ICA analysis. CI artifacts, as are incorporated in the continuous processing of sound stimuli from sound onset, are temporally aligned to EEG recordings.

Application of second-order blind identification (SOBI) through EEGLAB for each participant trial of continuous EEG data, will assist in visual topographical identifications of CI artifact components, with accordance to the implant side used during the listening paradigm. SOBI, a blind source separation algorithm, decomposes high density EEG data to recover stimulated neuronal sources spatially and temporally consistent with noise, improve signal-to-noise ratio (SNR) of evoked-potentials and furthermore reduce the level of subjectivity implicating source localization processes (Tang et al., 2005). Statistical analysis of mean and standard deviation of components corresponding to artifact topographies removed will be correlated for each participant. In order to prepare for alpha oscillation time-frequency analysis and temporal response functions (TRF), the cleaned EEG and aligned audio stimuli will be imported into EEGLAB and re-referenced to an average reference of all EEG sensors before further processing in EEGLAB.

Data Analysis:

Temporal Response Functions:

To assess cortical tracking, a linear forward modeling temporal response analysis will be applied to estimate the correspondence between attended audiovisual and ignored babble speech audio envelopes in correspondence to the continuous neural responses. Temporal response functions (TRF's) will be processed in MATLAB 2023b using a linear forward modeling. Weights across a range of time lags characterizes the functions, indicating, on a temporal basis, magnitudes of neural response changes to changes in audio stimuli. A TRF weight with a time lag of +200 ms

determines the direction and degree of change in the EEG signal 200 ms after a change in the stimulus envelope. Literature has indicated a delay in cortical tracking in CI patients at 150 ms compared to the generalized cortical tracking peak at 100 ms (Paul et al., 2020). Furthermore, we have selected to compare TRF weights from -100 ms to 500 ms.

To disentangle and assess neural responses in response to individual multifaceted features of attended, and ignored, audio speech input, multivariate TRF's (mTRFS) will be estimated. We will be using the mTRF toolbox and TRF and mTRF functions on MATLAB 2023 as presented by (Crosse et al., 2016). We will estimate mTRF functions over an integration window of -100 ms to 500 ms using a ridge regression coefficient of 2^{10}. The speech envelope is fitted to the neural signal using ridge regression, which uses a regularization coefficient parameter that reduces overfitting. Regularization parameters may be selected specific to participant trial data or an optimized fixed regularization can be applied across all participant trials.We have selected to optimize a regularization parameter to be used across all trials using the mTRFcrossval function. mTRFcrossval uses an N-fold cross validation, in which TRF's are estimated for each individual across all apart from N trials. The resulting TRF model predicts the EEG response to the audio envelope of the omitted trial and with Pearson's r correlation we can assess the degree of agreement between the mTRFcrossval predicted EEG and actual trial EEG recordings. The coefficients will be averaged across all participant trials within each condition and the value of maximum correspondence will be selected as the optimized regularization parameter. With the optimized regression coefficient, continuous EEG recordings will be epoched into 60 second trials for TRF analysis. TRF epochs will be averaged across the three trials within each condition for every participant. Furthermore, we will generate one TRF per condition for every participant to perform within-subject and between subject condition analysis. We will extract the absolute magnitude of TRF neural responses over a frontocentral scalp ROI, encompassing the participant based digitized EEG sensors (Figure 2).

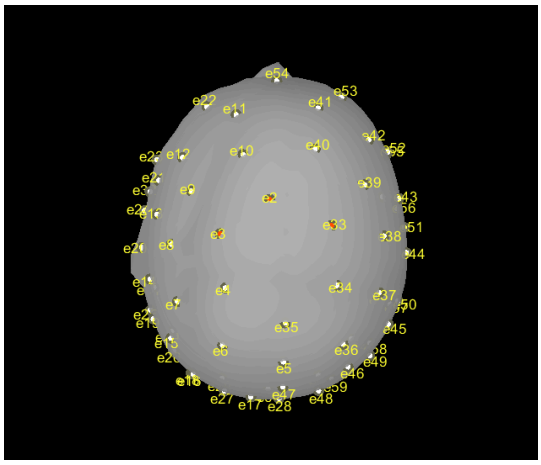


Figure 2. Region of interest for TRF analysis. The three frontocentral EEG sensors, as selected in red. Coordinates range based on participant based Polhemus electrode digitalization.

Brain Source Analysis of Alpha Power Lateralization:

Utilizing source localization, we will identify and compare between-subject analysis of region specific neural alpha-band modulation in response to each sensory condition. Previously, we demonstrated a correlation between speech perception and listening effort as a correlation of neural tracking and relative alpha oscillatory power measurement within the left Inferior Frontal cortex and associated Perisylvian Cortex Areas, particularly, Broca's Area (Dimitrijevic et al., 2019) (Bowen Xiu et al., 2022). Literature has also suggested that parietal alpha oscillations may reflect cross-modal attentional modulation (e.g., Bauer et al., 2020). Brainstorm software will be used to extract relative alpha power from ICA-processed EEG data after TRF analysis, following the brainstorm pipeline analysis methodology of (Niso et al., 2019). Source estimates of power and localization of neural responses will be computed using Brainstorms imaging kernel: standardized low-resolution electromagnetic tomography (sLORETA). sLORETA was selected according to its relatively high reputation for optimizing localization of sources, even when deep, with zero localization error (Pascual-Marqui, n.d.). At a spatial resolution of 5 mm, sLORETA divides the gray matter of the cortex into 6239 voxels (McCracken et al., 2022). Accordingly, to generate sLORETA models, boundary element models (BEM) will be created using the OpenMEEG Brainstorm plugin applying previously registered subject specific Polhemus EEG coordinates.

Through processing of sLORETA models of each participant, we will derive power spectral density (PSDs) estimates of sLORETA constructed voxels using Welch's method on Brainstorm. Prior to alpha specific PSDs, to attest ICA eliminated EEG signal variance associated with electrophysiological and cochlear implant interference, we will review frequency content across all source locations and time series with power spectral density (PSD) estimates and sensor topography of signal traces from 0 to 125 Hz. Further, by adjusting algorithmic input on Welch's model, PSDs will be estimated from 8 to 12 Hz using a sliding hamming window of 3 seconds at 50% overlap to localize modulation of alpha frequencies across trials. Spectrum normalization will then be processed, with a computation of RelativePSD(f) = PSD(f)/Σi[Total PSD(fi)], calculating at each source location and for each frequency band the ratio of the power contributed by the signal in that band to the total power of the source (Niso et al., 2019). Derived source and time series of relative alpha power spectral densities will be averaged for each condition across all participants for subsequent processing of ROI's. To localize alpha power and expected lateralization we will extract relative magnitude of alpha power over left and right ROI's within perisylvian cortex areas. Specifically, ROI's of the transverse temporal cortex, a $9.68cm^2$ region, and the superior temporal gyrus, a $102.21cm^2$ region as defined by the Desikan-Killiany Atlas will be evaluated, encompassing a total $144.48cm^2$ perisylvian area

(Desikan et al., 2006) (Figure 3). In correspondence to additional literature, parieto-occipital cortex regions will be analyzed as associative sensory processing and integration areas.

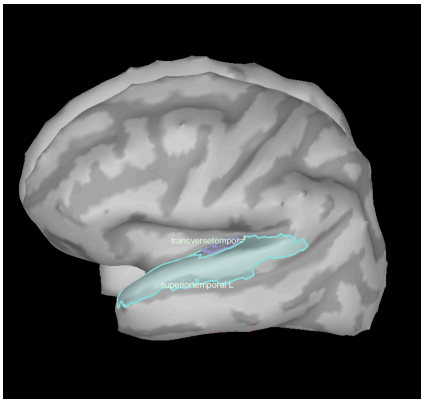


Figure 3. Regions of interest. Transverse temporal cortex, $9.68 cm^2$, as highlighted in purple. Superior temporal gyrus, $102.21 cm^2$, as highlighted in blue.

Preliminary Supporting Research:

Provided analysis of a pilot execution of the experimental design elicits the results we expect to obtain. The participant, male and 13 years of age, met within the inclusion criteria of the normal hearing group, furthermore the following results demonstrate selective auditory processing in typical development. Randomization of three trials per condition, subjective cognitive demand and comprehension scores are illustrated in figure 4.

ORDER	Condition	Attention Effort (1-10)	Did they give up?	Comprehension Q's (# = correct/4)
1	Quiet 1	1	No	4
2	Colocated 1	2	No	4
3	Right 1	4	No	4
4	Left 1	5	No	4
5	Quiet 2	2	No	4
6	Colocated 2	5	No	4
7	Right 2	6	No	4
8	Left 2	5	No	4
9	Quiet 3	1	No	4
10	Colocated 3	4	No	4
11	Right 3	6	No	4
12	Left 3	5	No	4

Figure 4. Participant Experiment Summary Sheet.

Temporal Response Functions and Brain Source Analysis:

Following the preprocessing procedures outlined in the methodology section above, we offline aligned audio speech envelopes of soundtrack, band-pass filtered, downsampled and removed technical and physiological EEG artifacts from raw participant EEG recordings. Provided temporal response methods were applied to processed EEG recordings. An average of 7.3 components were removed from participant trial data (11.6%, SD = 2.10). Results, figure 5, illustrate variations in neural responses to quiet, right, left or co-located multi-babble talk conditions.

(A) (B)

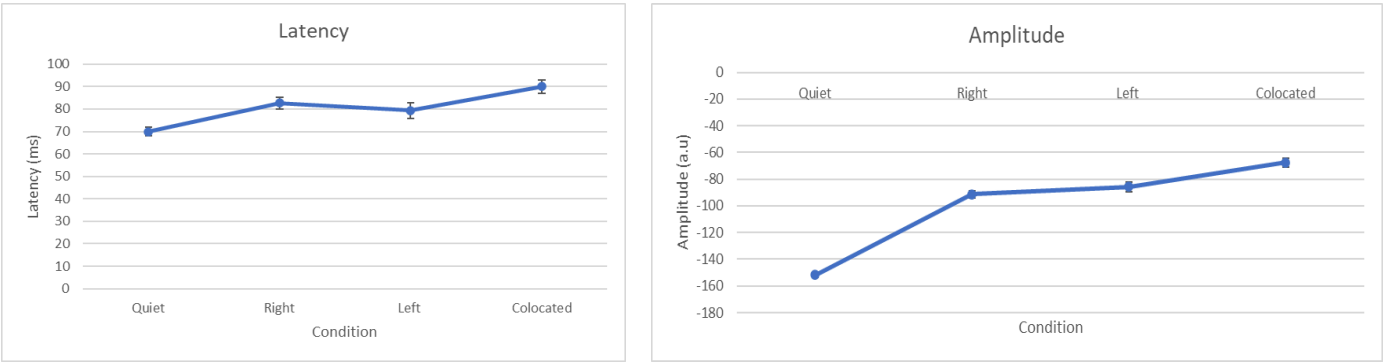


Figure 5. TRF Analysis: (A) Latency of Peak TRF (ms) and (B) Amplitude of Peak TRF (a.u)
Average latency of peak TRF (Quiet (M = 70, SD = 2), Right (M = 82.7, SD = 2.52), Left (M = 79.3, SD = 3.51), Co-located (M = 90.0, SD = 3.00)) and average amplitude of peak TRF (Quiet (M = -151.667, SD = 2.52), Right (M = -91.3, SD = 3.21), Left (M = -85.7, SD = 3.06), Co-located (M = -67.7, SD = 2.52)) was calculated across conditions.

We found baseline, at quiet, the negative evoked neural potential (N100) peak, indicative of stimulus onset and registry, in frontocentral ROI's was at 70 ms, although 80-100 ms is typical of adults, is of normal latency in children. In conditions of higher perceptual load, temporal phase locking of neural frequencies to speech envelope of speech stimulus (soundtrack) was delayed, significantly provided by a latency difference of 20 ms latency between the quiet condition and co-located condition. Amplitude, strongly dependent on temporal stimulus onset, loudness, frequency and to surrounding sound sources, is significantly decreased by high perceptual loads of surrounding sound. Topographies of averaged TRF's were processed corresponding to the digitized participant electrode locations (Figure 6). Amplitude peak was of greatest magnitude and more localized in the frontocentral region in conditions of quiet. Alpha oscillatory localization source analysis however was not performed for this set of participant trials.

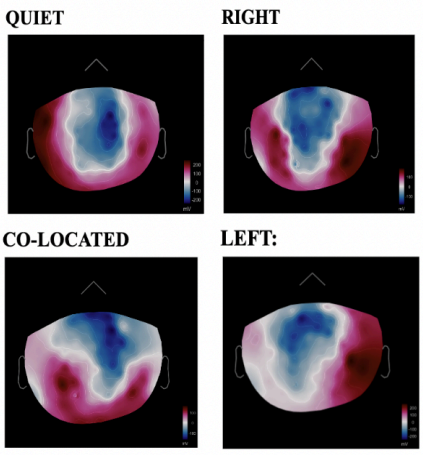


Figure 6. Topographies of Averaged TRF's
Topographies of TRF negative peak (blue).

Conclusion:

Furthermore, we consider how top-down processes and frequency phase locking (neural tracking) work as individual and integrative mechanisms to facilitate speech processing in a flexible manner. We focus on environments with a high perceptual load in which cognitive resources acquire extensive taxing to decipher variant sound sources and pertain attention to target stimuli. Our analysis will result with individual modulatory effects of alpha frequency band lateralization and neural entrainment (TRF's) to spatial allocation of irrelevant distracting stimuli. We will correlate the magnitude of relationship between mechanisms and their correspondence to self perceived measures of cognitive demand and comprehension and working memory capacity. With analysis and comparison of children of typical development and those of sensorineural hearing loss with cochlear implantation we aim to highlight fruitful directions for further research, prosper understanding of neurophysiological mechanisms to sensory processing and direct technological advancements to cochlear implantation thereby increasing the quality of life of those with hearing impairments.

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